

Bitácora

Instalación de Docker, PostgreSQL y DBeaver

Andrea Lucia Olmos Ortiz
Fundamentos de Bases de Datos

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Sistema operativo y versión: macOS Sonoma 14.5, chip Apple M1 (arquitectura ARM64)

Distribución (solo en caso de Linux): No aplica

Tiempo requerido: 53 minutos

Versión de la instalación

- **Docker Desktop:** Docker version 29.7.2, build a7dcaa6
- **PostgreSQL:**

```
version
-----
PostgreSQL 18.6 (Debian 18.6-1.pgdg13+2) on aarch64-unknown-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
(1 row)
```

- **DBeaver:** Version 26.2.0.20260830173

Explicación paso a paso

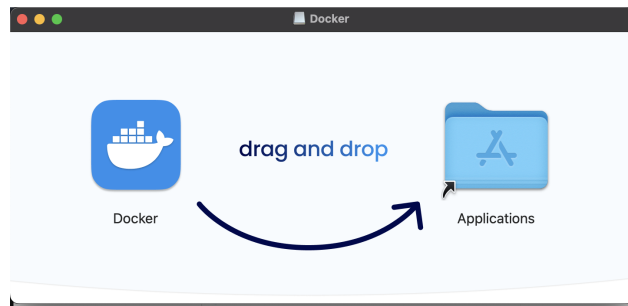
Instalación de Docker

1. Descargamos Docker en la link proporcionado en el pdf





2. Una vez descargado, damos arrastramos Docker a la carpeta de aplicaciones.



3. Comprobamos que se instaló correctamente usando el siguiente comando

```
docker run hello-world
```

```
[andy_1_olmos@AndyLap ~]()$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
58dee6a49ef1: Pull complete
c3bdf82c34d1: Download complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (arm64v8)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

Instalación de PostgreSQL

1. Descargamos la imagen de postgres y creamos el contenedor con los siguientes comandos.

```
docker pull postgres
```

```
[andy_1_olmos@AndyLap ~]$ docker pull postgres
Using default tag: latest
latest: Pulling from library/postgres
93eeac19656f: Pull complete
dc0e071a4d07: Pull complete
4f28b201a80c: Pull complete
4e582dc1d360: Pull complete
ab1e9043b37f: Pull complete
a3089dfdc569: Pull complete
521aece6b33d: Pull complete
bc5ea3235db8: Pull complete
bf7af0229701: Pull complete
6437d6ec8fee: Pull complete
ebe3b74b62b8: Pull complete
1a59b8215a09: Pull complete
3e1f39995a92: Pull complete
f833d8542db1: Download complete
4e9eb4879291: Download complete
Digest: sha256:4ef4dbc939d61acea57712655ddb4b4ab27419c913f94cca0cd57cb3ea3c2280
Status: Downloaded newer image for postgres:latest
docker.io/library/postgres:latest
```

```
docker run -d \
--name postgres \
-e POSTGRES_PASSWORD=mysecretpassword \
-p 127.0.0.1:5432:5432 \
-v postgres_data:/var/lib/postgresql \
postgres
```

Si necesitáramos inicializarlo una vez ya creado y este no está activo tenemos que averiguar el CONTAINER ID de PostgreSQL con el comando:

```
docker ps -a
```

1. Iniciamos el conetenedor

```
docker start 4d77b2f26540
```

```
[andy_1_olmos@AndyLap ~]$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
93c9d9f30b85	postgres	"docker-entrypoint.s..."	4 minutes ago	Exited (2) About a minute ago		gracious_yalow
3663596e23d8	postgres	"docker-entrypoint.s..."	4 minutes ago	Exited (2) 4 minutes ago		vibrant_williamson
344f717c8043	postgres	"docker-entrypoint.s..."	5 minutes ago	Exited (0) 4 minutes ago		inspiring_payne
bf05ac1a76fd	postgres	"docker-entrypoint.s..."	6 minutes ago	Exited (2) 5 minutes ago		festive_keller
b566fe50b6b7	postgres	"docker-entrypoint.s..."	6 minutes ago	Exited (2) 6 minutes ago		ecstatic_thompson
ec88bcfdadaa	postgres	"docker-entrypoint.s..."	6 minutes ago	Exited (2) 6 minutes ago		infallible_tesla
c329d3bbbbb3	postgres	"docker-entrypoint.s..."	8 minutes ago	Created		zen_villani
4d77b2f26540	postgres	"docker-entrypoint.s..."	15 minutes ago	Exited (0) About a minute ago		postgres
aa4e82e58654	hello-world	"/hello"	2 hours ago	Exited (0) 2 hours ago		focused_swartz
353152f90303	docker/welcome-to-docker:latest	"/docker-entrypoint..."	32 hours ago	Exited (0) 3 hours ago		welcome-to-docker

```
[andy_1_olmos@AndyLap ~]$ docker start 4d77b2f26540
4d77b2f26540
```

2. Verificamos que el contenedor está corriendo

```
docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
```

```
[andy_1_olmos@AndyLap ~]()$ docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
WARNING: Links on the default bridge network are deprecated and will be removed in a future release. Use a custom network instead.
Password for user postgres:
psql (18.6 (Debian 18.6-1.pgdg13+2))
Type "help" for help.

postgres=# hola
postgres=# \q
```

3. Detenemos el contenedor

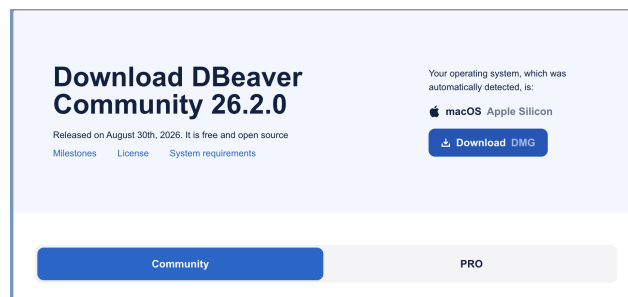
```
docker stop 4d77b2f26540
```

```
[andy_1_olmos@AndyLap ~]()$ docker stop 4d77b2f26540
4d77b2f26540
```

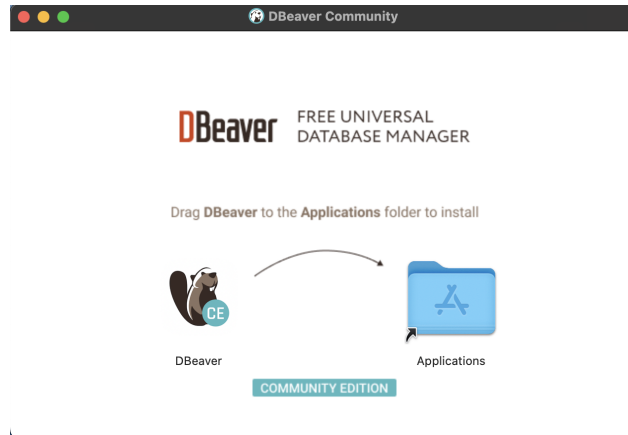
Conexión con DBeaver

1. Instalación de DBeaver Community Edition.

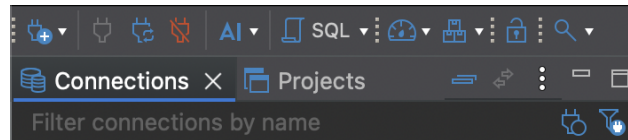
- a) Descargamos DBeaver en la link proporcionado en el pdf



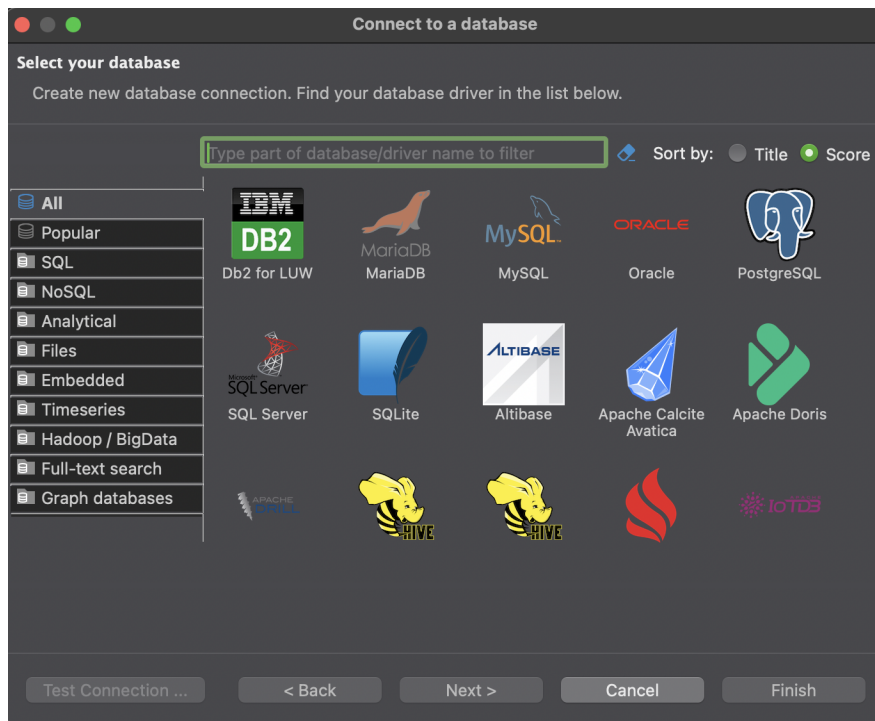
b) Un vez descargado, arrastramos la carpeta a Aplicaciones



2. Le damos clic al ícono de la clavija+ en la esquina superior izquierda



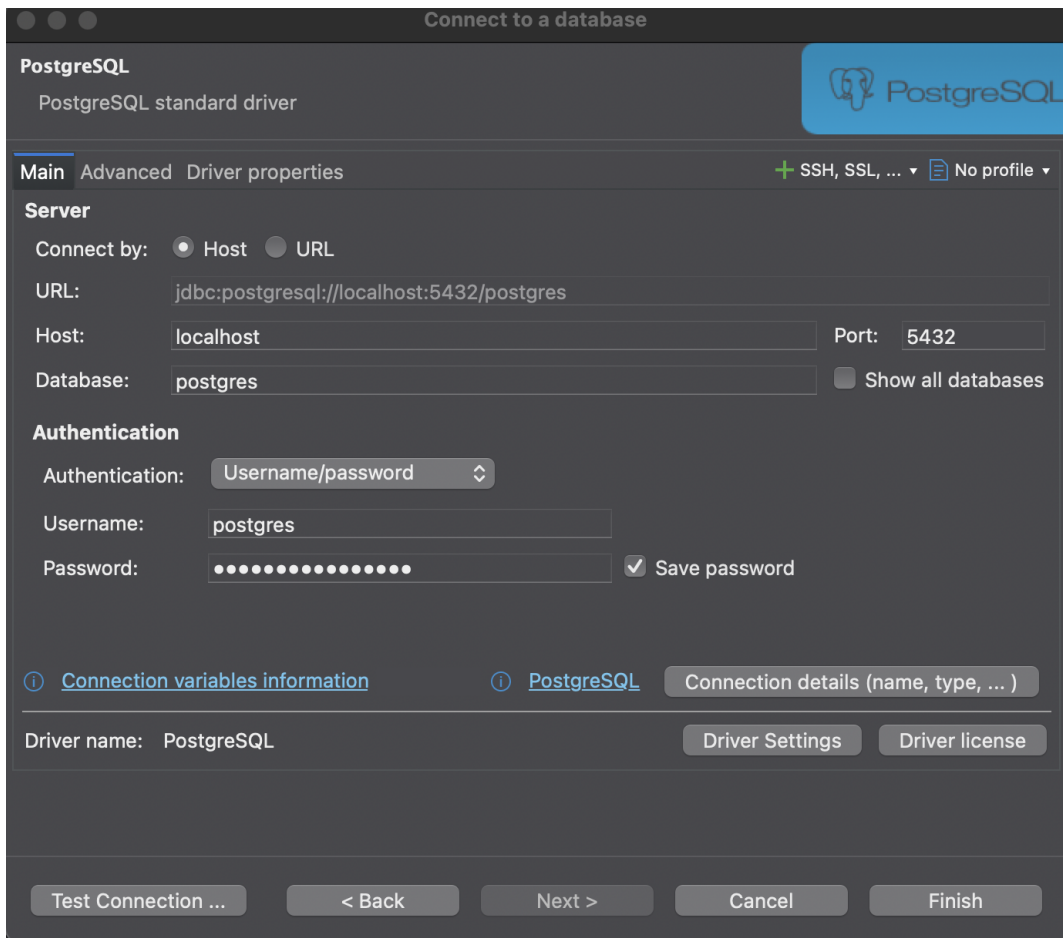
3. Seleccionamos PostgreSQL



4. Creamos la conexión con PostgreSQL.

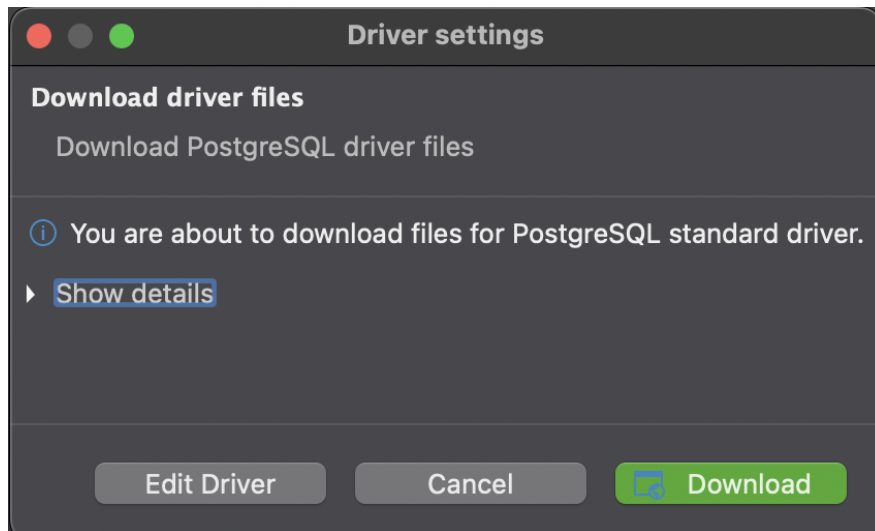
Parámetros de conexión:

- Host: localhost
- Puerto: 5432
- Base de datos: postgres
- Usuario: postgres
- Contraseña: Definida en POSTGRES_PASSWORD

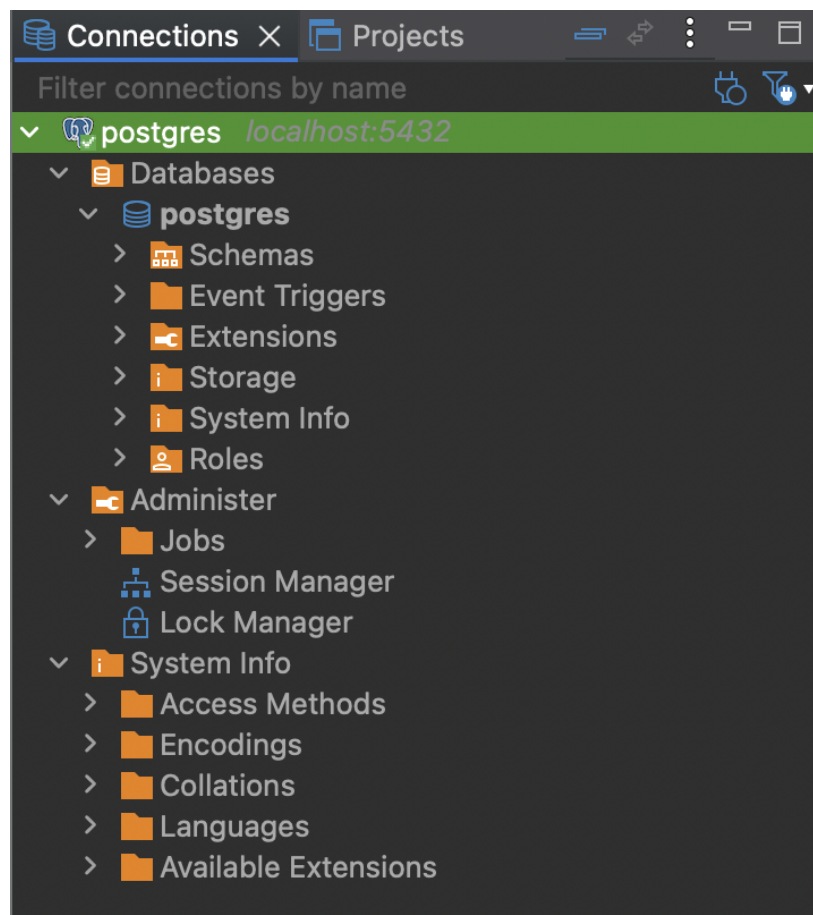


The screenshot shows a 'Connect to a database' dialog box with a dark theme. The title bar says 'Connect to a database'. The main header is 'PostgreSQL' with a PostgreSQL logo on the right. Below the header, it says 'PostgreSQL standard driver'. There are three tabs: 'Main' (selected), 'Advanced', and 'Driver properties'. To the right of the tabs are two buttons: '+ SSH, SSL, ...' and 'No profile'. The 'Server' section has 'Connect by:' with 'Host' selected and 'URL' unselected. Below this are fields for 'URL:' (jdbc:postgresql://localhost:5432/postgres), 'Host:' (localhost), 'Port:' (5432), and 'Database:' (postgres). There is a checkbox 'Show all databases' which is unchecked. The 'Authentication' section has a dropdown 'Authentication:' set to 'Username/password'. Below are fields for 'Username:' (postgres) and 'Password:' (masked with dots). There is a checkbox 'Save password' which is checked. At the bottom of the main area, there are two links: 'Connection variables information' and 'PostgreSQL', followed by a button 'Connection details (name, type, ...)'. Below this is a section for 'Driver name: PostgreSQL' with buttons 'Driver Settings' and 'Driver license'. At the very bottom, there are five buttons: 'Test Connection ...', '< Back', 'Next >', 'Cancel', and 'Finish'.

5. DBeaver pide descargar el driver JDBC. Le damos click en 'Download' y esperamos a que termine.



6. Finalmente, tenemos Postgres como conexión



Comentarios y problemas enfrentados

La instalación de Docker y DBeaver resultó bastante sencilla y directa. Lo que me pareció más complicado fue PostgreSQL, no tanto en el proceso de instalación en sí, ya que copié y pegué los comandos del PDF sin mayor problema, pero al intentar iniciarlo una vez ya creado y probar la consola (psql) el contenedor me salía este error:

```
docker: Error response from daemon: Cannot link to a non running container
: /postgres AS /zen_villani/postgres
```

Lo solucioné poniendo los comandos en el orden correcto (intenté conectarme mientras el contenedor estaba detenido), pero creo que me habría tardado menos si hubiera tenido más idea de qué son los contenedores y por qué los estamos usando para esto.

Al principio también tuve dudas sobre cómo conectar DBeaver con el contenedor de PostgreSQL, pero una vez que identifiqué el ícono de "nueva conexión", el proceso resultó muy intuitivo.